

Homework 10

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2.7.4

It is trivial that

$$\{\bar{x}, \bar{x}\} = \{\bar{y}, \bar{y}\} = \{\bar{p}_x, \bar{p}_x\} = \{\bar{p}_y, \bar{p}_y\} = 0$$

$$\{\bar{x}, \bar{y}\} = \frac{\partial \bar{x}}{\partial x} \frac{\partial \bar{y}}{\partial p_x} - \frac{\partial \bar{x}}{\partial p_x} \frac{\partial \bar{y}}{\partial x} + \frac{\partial \bar{x}}{\partial y} \frac{\partial \bar{y}}{\partial p_y} - \frac{\partial \bar{x}}{\partial p_y} \frac{\partial \bar{y}}{\partial y} = 0$$

$$\{\bar{x}, \bar{p}_x\} = \frac{\partial \bar{x}}{\partial x} \frac{\partial \bar{p}_x}{\partial p_x} - \frac{\partial \bar{x}}{\partial p_x} \frac{\partial \bar{p}_x}{\partial x} + \frac{\partial \bar{x}}{\partial y} \frac{\partial \bar{p}_x}{\partial p_y} - \frac{\partial \bar{x}}{\partial p_y} \frac{\partial \bar{p}_x}{\partial y} = \cos^2 \theta + \sin^2 \theta = 1$$

$$\{\bar{x}, \bar{p}_y\} = \frac{\partial \bar{x}}{\partial x} \frac{\partial \bar{p}_y}{\partial p_x} - \frac{\partial \bar{x}}{\partial p_x} \frac{\partial \bar{p}_y}{\partial x} + \frac{\partial \bar{x}}{\partial y} \frac{\partial \bar{p}_y}{\partial p_y} - \frac{\partial \bar{x}}{\partial p_y} \frac{\partial \bar{p}_y}{\partial y} = \cos \theta \sin \theta - \sin \theta \cos \theta = 0$$

$$\{\bar{y}, \bar{p}_x\} = \frac{\partial \bar{y}}{\partial x} \frac{\partial \bar{p}_x}{\partial p_x} - \frac{\partial \bar{y}}{\partial p_x} \frac{\partial \bar{p}_x}{\partial x} + \frac{\partial \bar{y}}{\partial y} \frac{\partial \bar{p}_x}{\partial p_y} - \frac{\partial \bar{y}}{\partial p_y} \frac{\partial \bar{p}_x}{\partial y} = \sin \theta \cos \theta - \cos \theta \sin \theta = 0$$

$$\{\bar{y}, \bar{p}_y\} = \frac{\partial \bar{y}}{\partial x} \frac{\partial \bar{p}_y}{\partial p_x} - \frac{\partial \bar{y}}{\partial p_x} \frac{\partial \bar{p}_y}{\partial x} + \frac{\partial \bar{y}}{\partial y} \frac{\partial \bar{p}_y}{\partial p_y} - \frac{\partial \bar{y}}{\partial p_y} \frac{\partial \bar{p}_y}{\partial y} = \sin^2 \theta + \cos^2 \theta = 1$$

$$\{\bar{p}_x, \bar{p}_y\} = \frac{\partial \bar{p}_x}{\partial x} \frac{\partial \bar{p}_y}{\partial p_x} - \frac{\partial \bar{p}_x}{\partial p_x} \frac{\partial \bar{p}_y}{\partial x} + \frac{\partial \bar{p}_x}{\partial y} \frac{\partial \bar{p}_y}{\partial p_y} - \frac{\partial \bar{p}_x}{\partial p_y} \frac{\partial \bar{p}_y}{\partial y} = 0$$

Therefore, the Poisson Brackets are preserved, and the transformation is a canonical transformation.

2.8.1

$$\bar{q}_1 = q_1 + \epsilon \frac{\partial p}{\partial p_1} = q_1 + \epsilon$$

$$\bar{q}_2 = q_2 + \epsilon \frac{\partial p}{\partial p_2} = q_2 + \epsilon$$

$$\bar{p}_1 = p_1 - \epsilon \frac{\partial p}{\partial q_1} = p_1$$

$$\bar{p}_2 = p_2 - \epsilon \frac{\partial p}{\partial q_2} = p_2$$

Therefore, $\varphi = p_1 + p_2$ is the generator of infinitesimal translations for a two-particle system.

2.8.2

$$\bar{q}_i = q_i + \epsilon \frac{\partial q}{\partial p_i}$$

$$\bar{p}_j = p_j - \epsilon \frac{\partial q}{\partial q_j}$$

$$\{\bar{q}_i, \bar{p}_j\} = \sum_k \frac{\partial \bar{q}_i}{\partial q_k} \frac{\partial \bar{p}_j}{\partial p_k} - \frac{\partial \bar{q}_i}{\partial p_k} \frac{\partial \bar{p}_j}{\partial q_k}$$

$$= \sum_k \left(\delta_{ik} + \epsilon \frac{\partial^2 q}{\partial p_i \partial q_k} \right) \left(\delta_{jk} - \epsilon \frac{\partial^2 q}{\partial q_j \partial p_k} \right) - \epsilon \frac{\partial^2 q}{\partial p_i \partial p_k} \cdot \epsilon \frac{\partial^2 q}{\partial q_i \partial q_k}$$

$$\begin{aligned}
&= \sum_k \delta_{ik} \delta_{jk} + \epsilon \frac{\partial^2 g}{\partial p_i \partial p_k} \cdot \delta_{jk} - \delta_{ik} \frac{\partial^2 g}{\partial p_j \partial p_k} + O(\epsilon^2) \\
&= \delta_{ij} + \epsilon \frac{\partial^2 g}{\partial p_i \partial p_j} - \epsilon \frac{\partial^2 g}{\partial p_j \partial p_i} + O(\epsilon^2) \\
&= \delta_{ij} + O(\epsilon^2) \\
&\approx \delta_{ij}
\end{aligned}$$

Therefore, the poisson brackets are preserved and g is a canonical transformation.

2.8.3

Rotation of just the coordinates:

$$\begin{aligned}
\begin{cases} \bar{x} = x \cos \theta - y \sin \theta \\ \bar{y} = x \sin \theta + y \cos \theta \end{cases} & \begin{cases} \bar{p}_x = p_x \\ \bar{p}_y = p_y \end{cases} \\
\{\bar{x}, \bar{y}\} = \frac{\partial \bar{x}}{\partial x} \frac{\partial \bar{y}}{\partial p_x} - \frac{\partial \bar{x}}{\partial p_x} \frac{\partial \bar{y}}{\partial x} + \frac{\partial \bar{x}}{\partial y} \frac{\partial \bar{y}}{\partial p_y} - \frac{\partial \bar{x}}{\partial p_y} \frac{\partial \bar{y}}{\partial y} &= 0 \\
\{\bar{p}_x, \bar{p}_y\} = \{p_x, p_y\} &= 0 \\
\{\bar{x}, \bar{p}_y\} = \frac{\partial \bar{x}}{\partial x} \frac{\partial \bar{p}_y}{\partial p_x} - \frac{\partial \bar{x}}{\partial p_x} \frac{\partial \bar{p}_y}{\partial x} + \frac{\partial \bar{x}}{\partial y} \frac{\partial \bar{p}_y}{\partial p_y} - \frac{\partial \bar{x}}{\partial p_y} \frac{\partial \bar{p}_y}{\partial y} &= -\sin \theta \neq 0 \\
\{\bar{x}, \bar{p}_x\} = \frac{\partial \bar{x}}{\partial x} \frac{\partial \bar{p}_x}{\partial p_x} - \frac{\partial \bar{x}}{\partial p_x} \frac{\partial \bar{p}_x}{\partial x} + \frac{\partial \bar{x}}{\partial y} \frac{\partial \bar{p}_x}{\partial p_y} - \frac{\partial \bar{x}}{\partial p_y} \frac{\partial \bar{p}_x}{\partial y} &= \cos \theta \neq 1 \\
\{\bar{y}, \bar{p}_x\} = \frac{\partial \bar{y}}{\partial x} \frac{\partial \bar{p}_x}{\partial p_x} - \frac{\partial \bar{y}}{\partial p_x} \frac{\partial \bar{p}_x}{\partial x} + \frac{\partial \bar{y}}{\partial y} \frac{\partial \bar{p}_x}{\partial p_y} - \frac{\partial \bar{y}}{\partial p_y} \frac{\partial \bar{p}_x}{\partial y} &= \sin \theta \neq 0 \\
\{\bar{y}, \bar{p}_y\} = \frac{\partial \bar{y}}{\partial x} \frac{\partial \bar{p}_y}{\partial p_x} - \frac{\partial \bar{y}}{\partial p_x} \frac{\partial \bar{p}_y}{\partial x} + \frac{\partial \bar{y}}{\partial y} \frac{\partial \bar{p}_y}{\partial p_y} - \frac{\partial \bar{y}}{\partial p_y} \frac{\partial \bar{p}_y}{\partial y} &= \cos \theta \neq 1
\end{aligned}$$

Therefore, some of the poisson brackets are not preserved, and the transformation is not a canonical transformation.

$$\begin{aligned}
\text{If } \delta H &= \epsilon \{H, g\} = \epsilon \left(\frac{\partial H}{\partial x} \frac{\partial g}{\partial p_x} - \frac{\partial H}{\partial p_x} \frac{\partial g}{\partial x} + \frac{\partial H}{\partial y} \frac{\partial g}{\partial p_y} - \frac{\partial H}{\partial p_y} \frac{\partial g}{\partial y} \right), \\
\text{we have } \delta x &= \epsilon \frac{\partial g}{\partial p_x}, \quad \delta p_x = -\epsilon \frac{\partial g}{\partial x}, \quad \delta y = \epsilon \frac{\partial g}{\partial p_y}, \quad \delta p_y = -\epsilon \frac{\partial g}{\partial y} \\
\Rightarrow \begin{cases} \bar{x} = x + \epsilon \frac{\partial g}{\partial p_x} \\ \bar{y} = y + \epsilon \frac{\partial g}{\partial p_y} \end{cases} & \begin{cases} \bar{p}_x = p_x - \epsilon \frac{\partial g}{\partial x} \\ \bar{p}_y = p_y - \epsilon \frac{\partial g}{\partial y} \end{cases}
\end{aligned}$$

According to 2.8.2, this is a canonical transformation. Therefore, there doesn't exist any g , such that $\delta H = \epsilon \{H, g\}$.

10.2.2

(1) We start with the following Hamiltonian:

$$H = \frac{p_x^2 + p_y^2}{2m} + \frac{1}{2} m \omega_x^2 x^2 + \frac{1}{2} m \omega_y^2 y^2$$

Introduce creation and annihilation operators as follows:

$$\begin{aligned} a_x &\equiv \frac{1}{\sqrt{2}} \left(\sqrt{\frac{m\omega_x}{\hbar}} x + i \sqrt{\frac{1}{m\hbar\omega_x}} p_x \right) \\ a_x^\dagger &\equiv \frac{1}{\sqrt{2}} \left(\sqrt{\frac{m\omega_x}{\hbar}} x - i \sqrt{\frac{1}{m\hbar\omega_x}} p_x \right) \\ a_y &\equiv \frac{1}{\sqrt{2}} \left(\sqrt{\frac{m\omega_y}{\hbar}} y + i \sqrt{\frac{1}{m\hbar\omega_y}} p_y \right) \\ a_y^\dagger &\equiv \frac{1}{\sqrt{2}} \left(\sqrt{\frac{m\omega_y}{\hbar}} y - i \sqrt{\frac{1}{m\hbar\omega_y}} p_y \right) \end{aligned}$$

Then we have that:

$$\begin{aligned} [a_x, a_y] &= [a_x, a_y^\dagger] = [a_x^\dagger, a_y] = [a_x^\dagger, a_y^\dagger] = 0 \\ [a_x, a_x^\dagger] &= \frac{1}{2} \frac{i}{\hbar} ([p, x] - [x, p]) \\ &= \frac{1}{2} \frac{i}{\hbar} \cdot -2i\hbar \\ &= 1 \\ [a_y, a_y^\dagger] &= 1, \text{ analogously} \end{aligned}$$

Furthermore, we have that

$$\begin{aligned} \hbar\omega_x \left(a_x^\dagger a_x + \frac{1}{2} \right) &= \frac{1}{2} \hbar\omega_x \left(\frac{m\omega_x}{\hbar} x^2 + \frac{1}{m\hbar\omega_x} p_x^2 - \frac{i}{\hbar} [p_x, x] + 1 \right) \\ &= \frac{1}{2} m\omega_x^2 x^2 + \frac{p_x^2}{2m} + \hbar\omega_x \left(-\frac{i}{\hbar} \cdot -i\hbar + 1 \right) \\ &= \frac{p_x^2}{2m} + \frac{1}{2} m\omega_x^2 x^2 \end{aligned}$$

Thus

$$H = \hbar\omega_x \left(a_x^\dagger a_x + \frac{1}{2} \right) + \hbar\omega_y \left(a_y^\dagger a_y + \frac{1}{2} \right)$$

We know the spectrum of a single harmonic oscillator. (If you don't remember, it comes from the fact that $a^\dagger a$ is a "number operator", that is, it has eigenvalues of all non-negative integers.) Using this and the fact that the x operators (that is, $a_x, a_x^\dagger$) and y operators commute, we have:

$$E = \hbar\omega_x \left(n_x + \frac{1}{2} \right) + \hbar\omega_y \left(n_y + \frac{1}{2} \right)$$

for non-negative integers n_x, n_y .

- (2) The combined harmonic oscillator will have a ground state wavefunction that is the product of ground state wavefunctions for an oscillator in the x direction (with frequency ω_x) and in the y direction (with frequency ω_y). This can be seen in an easy way: we know that there is exactly one ground state of the combined harmonic oscillator, with energy $E = \frac{1}{2}\hbar(\omega_x + \omega_y)$ so we attempt a solution of the time independent Schrödinger equation using a separable solution $\psi(x, y) = \psi_x(x)\psi_y(y)$. The time independent Schrödinger equation is exactly the

equation we derived previously without the modification due to oscillator coupling. We show how to solve this:

$$\underbrace{\left(-\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) + \frac{1}{2}m\omega_x^2 x^2 + \frac{1}{2}m\omega_y^2 y^2\right) \psi_x(x)\psi_y(y)}_{=f(x)} = \frac{1}{2}\hbar(\omega_x + \omega_y)\psi_x(x)\psi_y(y)$$

$$\underbrace{\frac{1}{\psi_x} \left(-\frac{\hbar^2}{2m} \frac{\partial^2 \psi_x}{\partial x^2} + \frac{1}{2}m\omega_x^2 x^2 \psi_x\right)}_{=f(x)} + \underbrace{\frac{1}{\psi_y} \left(-\frac{\hbar^2}{2m} \frac{\partial^2 \psi_y}{\partial y^2} + \frac{1}{2}m\omega_y^2 y^2 \psi_y\right)}_{=g(y)} = \frac{1}{2}\hbar(\omega_x + \omega_y)$$

thus the highlighted parts are each constants since we have equated a function of x only, a function of y only, and a constant. We are free to try the solution $f(x) = \frac{1}{2}\hbar\omega_x$, $g(y) = \frac{1}{2}\hbar\omega_y$, which produces the following equations:

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi_x}{dx^2} + \frac{1}{2}m\omega_x^2 x^2 \psi_x = \frac{1}{2}\hbar\omega_x \psi_x$$

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi_y}{dy^2} + \frac{1}{2}m\omega_y^2 y^2 \psi_y = \frac{1}{2}\hbar\omega_y \psi_y,$$

but we already know a solution to these! The solution is the harmonic oscillator ground state wavefunctions for x and y . Thus we have found a solution for ψ , and it satisfies the appropriate boundary conditions (we can set the normalization with a constant, and we have that $\psi \rightarrow 0$ as we approach infinity in any direction. Thus it is the solution for which we have been looking. We know furthermore that the state corresponding to n_x, n_y is $(a_x^\dagger)^{n_x} (a_y^\dagger)^{n_y} |0\rangle$, and since the $a_x^\dagger$ operators will only affect the x part of the wavefunction, and similarly for the $a_y^\dagger$ operators for the y part of the wavefunction, we have that $\psi_{n_x, n_y} = \psi_{n_x}(x)\psi_{n_y}(y)$, where ψ_{n_x} is the wavefunction for the n_x th energy eigenstate for a harmonic oscillator with frequency ω_x , and similarly for ψ_{n_y} (with frequency ω_y).

Under a parity transformation $x \rightarrow -x, y \rightarrow -y$, the wavefunction picks up a factor of $(-1)^{n_x}$ from changing $x \rightarrow -x$ in $\psi_{n_x}(x)$, and a similar factor of $(-1)^{n_y}$ from $\psi_{n_y}(y)$, so thus the overall factor is $(-1)^{n_x+n_y}$. Thus the eigenstates have definite parity, and that parity is $(-1)^{n_x+n_y}$ (where 1 indicates even under parity, -1 indicates odd under parity; this is standard notation, and if you read about the parity operator you'll see where it comes from).

- (3) For $\omega_x = \omega_y = \omega$, we know the general eigenstate has wavefunction:

$$\psi_{n_x, n_y}(x, y) = \left(\frac{m\omega}{\pi\hbar 2^{n_x+n_y} n_x! n_y!}\right)^{1/2} \exp\left(-\frac{m\omega(x^2 + y^2)}{2\hbar}\right) H_{n_x}\left[\left(\frac{m\omega}{\hbar}\right)^{1/2} x\right] H_{n_y}\left[\left(\frac{m\omega}{\hbar}\right)^{1/2} y\right]$$

We can easily write this down for the lowest three energy eigenstates, which correspond to $n_x = n_y = 0$ (lowest), and the two with $n_x = 1, n_y = 0$, $n_x = 0, n_y = 1$ (degenerate, next lowest):

$$\psi_{0,0}(x, y) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} \exp\left(-\frac{m\omega(x^2 + y^2)}{2\hbar}\right)$$

$$\psi_{1,0}(x, y) = \sqrt{\frac{2}{\pi}} \left(\frac{m\omega}{\hbar}\right)^{1/2} x \exp\left(-\frac{m\omega(x^2 + y^2)}{2\hbar}\right)$$

$$\psi_{0,1}(x, y) = \sqrt{\frac{2}{\pi}} \left(\frac{m\omega}{\hbar}\right)^{1/2} y \exp\left(-\frac{m\omega(x^2 + y^2)}{2\hbar}\right)$$

These are also easily re-expressable in terms of polar coordinates:

$$\begin{aligned}\psi_{0,0}(\rho, \phi) &= \left(\frac{m\omega}{\pi\hbar}\right)^{1/2} \exp\left(-\frac{m\omega\rho^2}{2\hbar}\right) \\ \psi_{1,0}(\rho, \phi) &= \sqrt{\frac{2}{\pi}} \left(\frac{m\omega}{\hbar}\right) \rho \cos(\phi) \exp\left(-\frac{m\omega\rho^2}{2\hbar}\right) \\ \psi_{0,1}(\rho, \phi) &= \sqrt{\frac{2}{\pi}} \left(\frac{m\omega}{\hbar}\right) \rho \sin(\phi) \exp\left(-\frac{m\omega\rho^2}{2\hbar}\right)\end{aligned}$$

We also have that the formula for the energy given n_x, n_y is

$$E = \hbar\omega(n_x + n_y + 1)$$

Thus the states with $n_x + n_y = n$ are all degenerate. So the state with $E = (n+1)\hbar\omega$ is $n+1$ degenerate: there are $n+1$ ways to choose $n_x + n_y = n$ if n_x, n_y are non-negative integers. (n_x can take any value 0 to n , hence we are not "off by 1").

10.3.3

- (1) The particles are to be considered distinguishable. Any number of particles can be in any one state.

So each of three particles A, B, C can be placed in any of one of three states. Hence there exist a total of $3^3 = 27$ possible states. $\Rightarrow$ The particles are said to obey "Maxwell-Boltzmann Statistics"

- (2) The particles are to be considered indistinguishable. This implies that $A = B = C$, so that the states in the case (1) which differed only in the interchange of particles are now no longer to be considered distinct. Hence there exist a total of 10 possible states. (Refer to the table below.)
- (3) The particles are to be considered indistinguishable. No more than one particle can be in any state. Hence there exists only one possible state.

Case (2)			Case (3)		
a	b	c	a	b	c
AAA			A	A	A
	AAA				
		AAA			
A	AA				
A		AA			
AA	A				
	A	AA			
	AA	A			
AA		A			
A	A	A			